

Concept 4 | Integration: Definite Integrals and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Classified Practice

Integration: Definite Integrals and Applications

Q001 Written

Find the two indefinite integrals in the Warm Up.

Q002 Written

Find the six indefinite integrals in the source Try section, expanding a squared bracket first when needed.

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Q003 **Written****Solve the first ten source indefinite-integral exercises (constants, linear and quadratic polynomials).**

Q004 Written

Solve the remaining eight source integrals, including products and squared brackets.

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Q005 Written

Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.

Q006 Written**Find the two source Try functions from their derivatives and one given value of F .**

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Q007 **Written****Find $F(x)$ for all six source Exercise conditions.**

Q008 Written**Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.**

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Q009 Written

Solve the source Try tangent-gradient problem through $(1, 1)$.

Q010 Written

Solve the three source tangent-gradient exercises, finding each function from its point and gradient.

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Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Q012 Written

Evaluate the six source Try definite integrals, simplifying the integrand before applying the limits.

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Q013 Written

Evaluate the first twelve source Exercise definite integrals, including reversed and equal limits.

Q014 Written

Evaluate the remaining twelve source Exercise integrals, including sums, differences and products.

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Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Q016 Written**Solve the four source Try definite-integral transformations.**

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Q017 **Written****Solve the first six source Exercise questions using interval splitting and reversal.**

Q018 Written

Solve the final six source Exercise questions using the factorized definite-integral identity.

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Q019 Written

Find $f(x)$ from the source equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Q020 Written**Solve the two source Try integral equations for $f(x)$.**

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Q021 Written

Find $f(x)$ for all five source integral equations.

Q022 Written**Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .**

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Q023 Written

Solve the two source Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Q024 Written**Differentiate the three source expressions with respect to x .**

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Q025 Written

Find $f(x)$ and a for the three source equations with a variable upper limit.

Q026 MCQ**The constant of integration is required because:****Choose the correct option.**

- A every primitive is unique
- B primitives differ by a constant
- C the integrand is constant
- D the limits are equal

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Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Choose the correct option.**

A $nx^{n-1} + C$

B $x^{n+1}/(n+1) + C$

C $x^n/n + C$

D $x^{n-1}/(n-1) + C$

Q028 MCQ**Before integrating $(x - 1)^2$, first:****Choose the correct option.**

- A differentiate it
- B expand it
- C set $x = 0$
- D add limits

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Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Choose the correct option.**

A $x^3 - x^2 + x + C$

B $3x^3 - 2x^2 + x + C$

C $x^3 - 2x + 1 + C$

D $6x - 2 + C$

Q030 MCQ**To find $F(x)$ from $F'(x)$, you should first:****Choose the correct option.**

- A differentiate again
- B integrate $F'(x)$
- C set $F = 0$
- D ignore the initial value

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Q031 MCQ

The condition $F(1) = 7$ determines:**Choose the correct option.**

- A the derivative
- B the domain
- C the constant C
- D the leading coefficient

Q032 MCQ**If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:****Choose the correct option.**

A $3x^2 - 2x + 1$

B $6x^2 - 2x + 1$

C $3x^2 - 2x$

D $6x - 2 + 1$

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Q033 MCQ

The gradient of the tangent at (x, y) is:**Choose the correct option.**A $f(x)$ B $f'(x)$ C $f''(x)$ D $x + y$

Q034 MCQ**A point (x_0, y_0) on $y = f(x)$ gives:****Choose the correct option.**

A $f'(x_0) = y_0$

B $f(x_0) = y_0$

C $f(0) = x_0$

D $C = 0$

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Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Choose the correct option.

A $3x^3 - 2x^2 + x$

B $x^3 - x^2 + x + C$

C $6x - 2$

D $x^3 - 2x + 1$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Choose the correct option.

A $F(a) - F(b)$

B $F(b) - F(a)$

C $F(a) + F(b)$

D 0 always

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Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Choose the correct option.

A 1

B a

C 0

D $f(a)$

Q038 MCQ

A reversed definite integral satisfies:**Choose the correct option.**

- A $\int_a^b f = \int_b^a f$
- B $\int_a^b f = -\int_b^a f$
- C both are zero
- D only if $f = 0$

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Q039 MCQ

To evaluate a definite integral of a product, first:**Choose the correct option.**

- A integrate each factor separately
- B expand the product
- C differentiate the product
- D change the limits

Q040 MCQ**Adjacent integrals with the same integrand can be:****Choose the correct option.**

- A multiplied
- B combined over the joined interval
- C set equal to zero
- D reversed only

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Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta)dx$, the factors must have:

Choose the correct option.

- A different roots
- B the displayed roots (a,b)
- C zero coefficients
- D equal limits

Q042 MCQ

The identity for the factored integral is:**Choose the correct option.**

A $(\beta - \alpha)^2/6$

B $-(\beta - \alpha)^3/6$

C $(\beta + \alpha)^3/6$

D 0

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Q043 MCQ

A minus sign before an integral can be removed by:**Choose the correct option.**

- A squaring the integrand
- B reversing its limits
- C adding C
- D differentiating

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Choose the correct option.

- A a function of x
- B a constant
- C always zero
- D a derivative

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Q045 MCQ

The best first step for an integral equation is to let:**Choose the correct option.**

A $A = f(x)$

B $A = \int_a^b f(t)dt$

C $A = x$

D $A = C$ only

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:

Choose the correct option.

A $1/3$

B $-1/3$

C 0

D 1

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Q047 MCQ

 $\frac{d}{dx} \int_a^x f(t)dt$ equals:

Choose the correct option.

A $f(a)$ B $f(x)$ C $F(x)$

D 0

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Choose the correct option.

- A the lower limit
- B the upper-limit variable
- C C
- D zero

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Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Choose the correct option.

- A the derivative
- B the unknown lower limit a
- C the integrand degree
- D the upper limit

Q050 MCQ**For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:****Choose the correct option.**

A $2x^2 + 3x - 5$

B $2x^3 + 3x^2 - 5x$

C $3x^2 + 3x - 5$

D 0

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Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Classified Practice

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Quiz WRITTEN 3 Concept Quiz · Written**Evaluate** $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

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Quiz MCQ 5 **Concept Quiz · MCQ****The derivative of $\int_0^x (3t^2 + 1)dt$ is:****Choose the correct option.**

A $3x^2 + 1$

B $x^3 + x$

C $6x$

D 0

Quiz MCQ 6 Concept Quiz · MCQ

A primitive of $5x^4$ is:

Choose the correct option.

A $x^5 + C$

B $5x^5 + C$

C $20x^3 + C$

D $x^4/5 + C$

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Quiz MCQ 7 [Concept Quiz](#) · [MCQ](#)

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Choose the correct option.

A $2x^2 + 2x + 3$

B $4x^2 + 2x + 3$

C $2x^2 + 2$

D $4x + 5$

Quiz MCQ 8 Concept Quiz · MCQ

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Choose the correct option.

- A $\int_1^5 f(x)dx$
- B $\int_1^3 f(x)dx$
- C $\int_3^5 f(x)dx$
- D 0

Answer key

Use the question number shown on each page.

Q001 Integrate term by term and add C to each primitive.	Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.	Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.	Q036 B
Q002 Expand products or powers, collect like powers, then integrate every term.	Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.	Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.	Q037 C
Q003 Apply linearity and the power rule; retain the constant of integration.	Q014 Use linearity, expand products, then apply the endpoint-substitution rule.	Q022 $2x^2 + 3x - 5$.	Q038 B
Q004 Expand each product or square, simplify, and integrate term by term.	Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.	Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a .	Q039 B
Q005 $F(x) = x^3 + 4x^2 - x + 3$.	Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.	Q024 Apply the variable-limit Fundamental Theorem directly to each integrand.	Q040 B
Q006 Integrate the derivative and use the given point to determine C .	Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits.	Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a .	Q041 B
Q007 For each item, integrate $F'(x)$, then impose the stated value of F .	Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$.	Q026 B	Q042 B
Q008 $f(x) = x^3 - x^2 + x - 19$.	Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.	Q027 B	Q043 B
Q009 Integrate the given gradient and use the point on the curve to determine the constant.		Q028 B	Q044 B
Q010 Integrate the polynomial gradient and apply the given point condition in each case.		Q029 A	Q045 B
Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.		Q030 B	Q046 B
		Q031 C	Q047 B
		Q032 A	Q048 B
		Q033 B	Q049 B
		Q034 B	Q050 A
		Q035 B	Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$
			Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$
			Quiz WRITTEN 3 6
			Quiz WRITTEN 4 $f(x) = x^2 - 1/3$
			Quiz MCQ 5 A
			Quiz MCQ 6 A
			Quiz MCQ 7 A
			Quiz MCQ 8 A